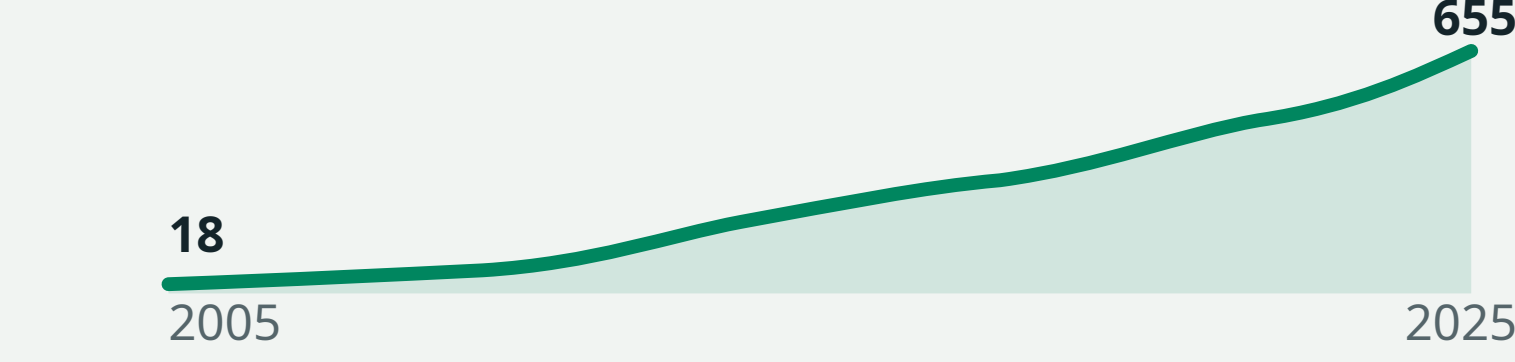


Perturbation evidence is multiplying faster than it can be organised

A single genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, while pooled screens test thousands of targets. That scale could reveal robust biology across studies—but identifiers, metadata, assays and processing pipelines differ. Results remain hard to compare or reproduce. The goal of the Perturbation Catalogue is to supply a shared schema, stable gene identity and one access path for combining this evidence.

36× growth
in perturbation papers
over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: Replogle et al., Cell 2022.

Our approach turns scattered datasets into a unified resource



We extensively curate metadata...

84 searchable fields in the dataset contract

Study & provenance

Title, authors, year, source
URI, licence, linked datasets

Perturbation & library

Targets, library design,
delivery, guides, variants

Model system

Organism, tissue, cell type or
line, disease, sex, stage

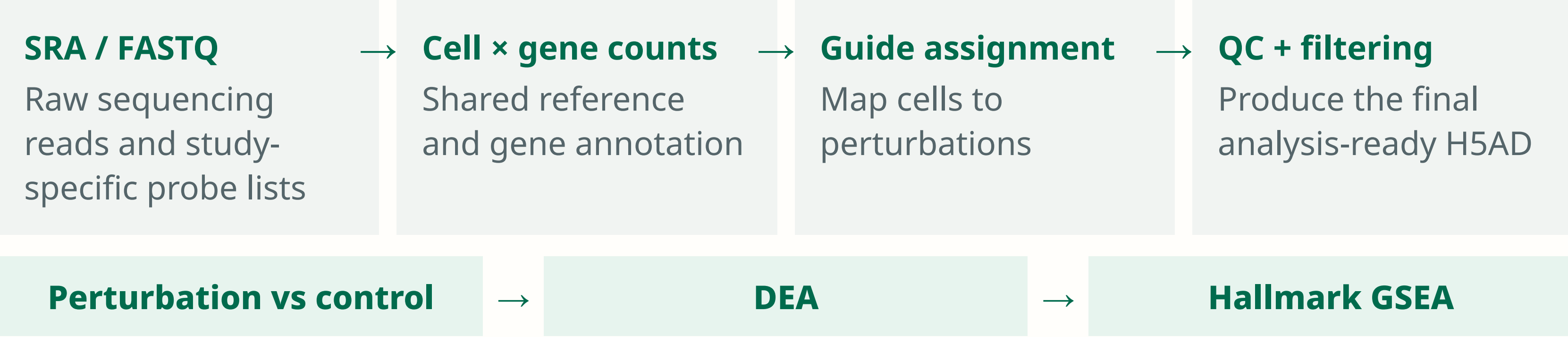
Readout & processing

Assay, technology,
sequencing kit, reference,
software

Stable Ensembl gene IDs and ontology-linked entities make the same biological question addressable across modalities.

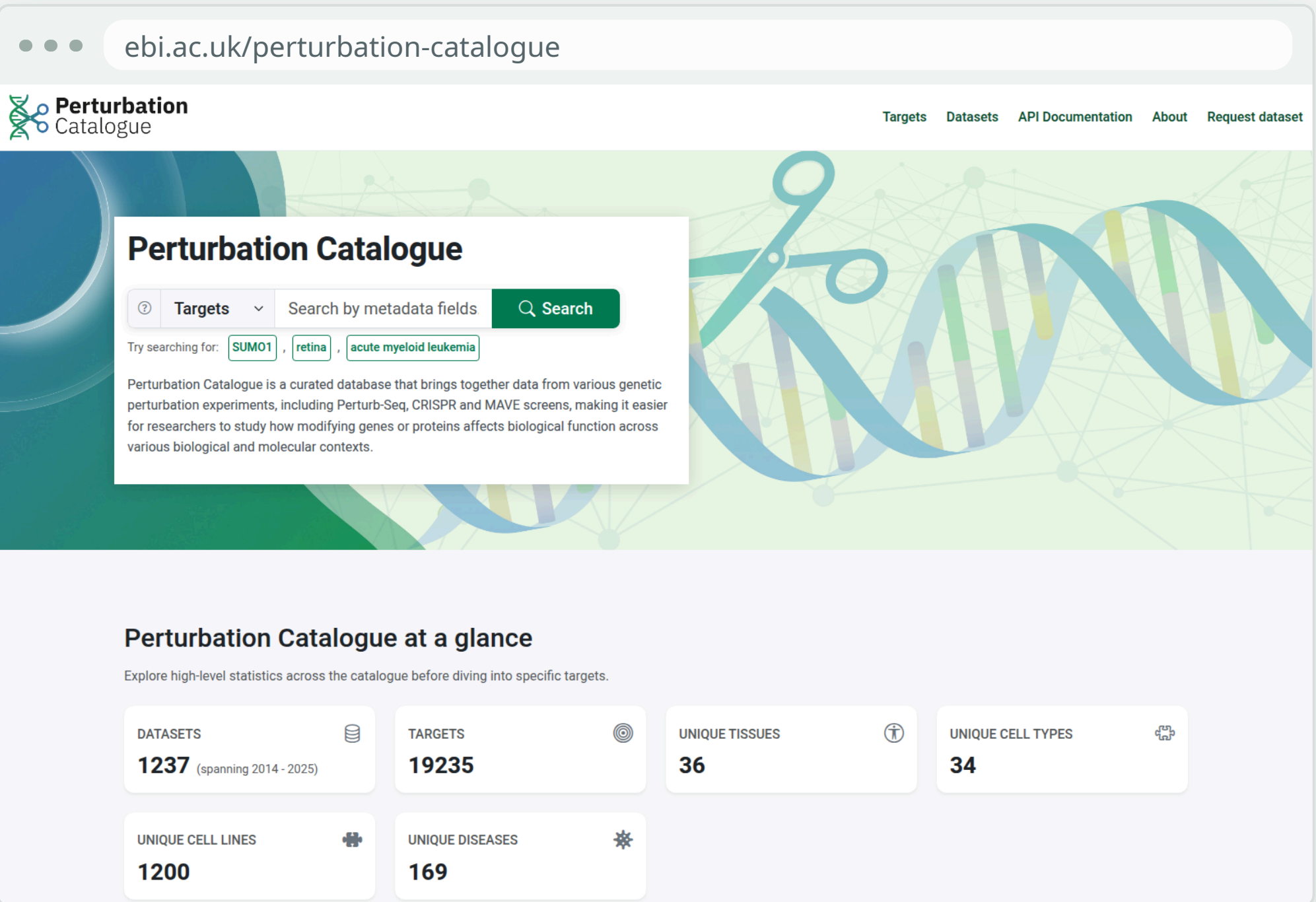
And reprocess data from scratch

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.



Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Browse the Catalogue



Target search, current coverage and biological-context summaries in one human-facing portal.

Access records via API

Search and retrieve metadata or modality results through the same backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
```

```
{
  "perturbation": {"gene_symbol": "TFAM"},
  "effect": {
    "gene_symbol": "MT-ND5",
    "log2fc": -1.926, "padj": 0.0
  }
}
```

Download whole datasets

Take complete modality tables with provenance and metadata for local analysis or archival use.

Parquet
Columnar, efficient for large-scale analysis

CSV.gz
Portable, compressed tabular export

JSON metadata
Machine-readable dataset context and provenance

More exciting things are coming soon

- ### Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.
- ### Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.
- ### Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.